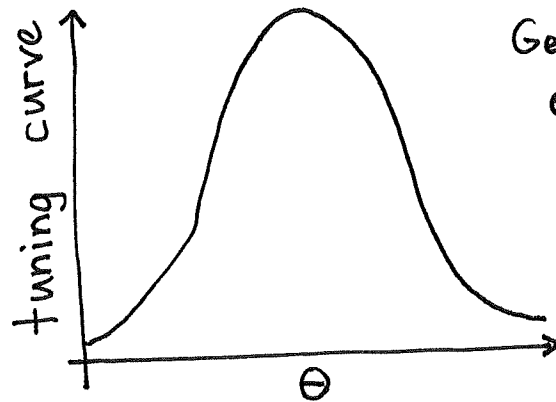
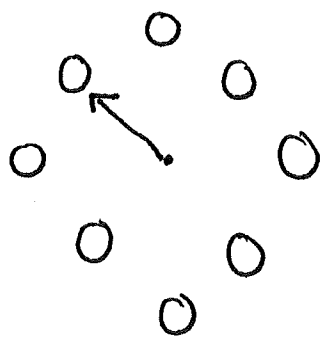


Low-dimensional dynamics in motor cortex:

Neural tuning in center out reach task:

Reaching task



Georgopoulos et. al

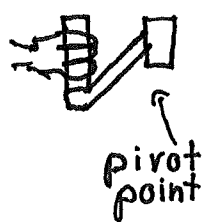
Monkeys repeatedly reach from central location to peripheral locations (typically 8 possibilities). Many neurons are tuned to position (e.g. reach direction), but other neurons do not have clear tuning for many movement parameter. Tuning may be epiphenomenal.

The more modern approach is to quantify how motor cortex dynamics facilitate the generation of body dynamics.

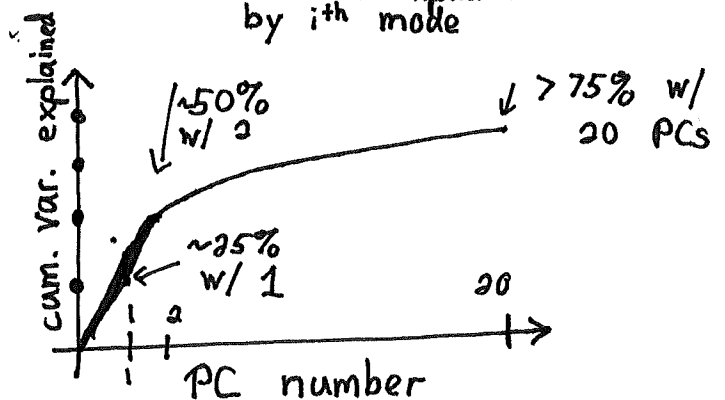
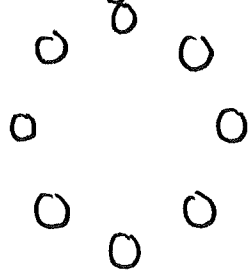
A critical observation is that neural activity is often low-dimensional. Variance decomposition:

$$\mathbf{r} = \sum_{i=1}^N \mathbf{P}_i \xi^{(i)} \Rightarrow \langle \mathbf{r}^T \mathbf{r} \rangle = \sum_{i=1}^N \underbrace{\langle \mathbf{P}_i^2 \rangle}_{\text{Variance explained by } i^{\text{th}} \text{ mode}} = \sum_{i=1}^N \lambda_i$$

Wrist task



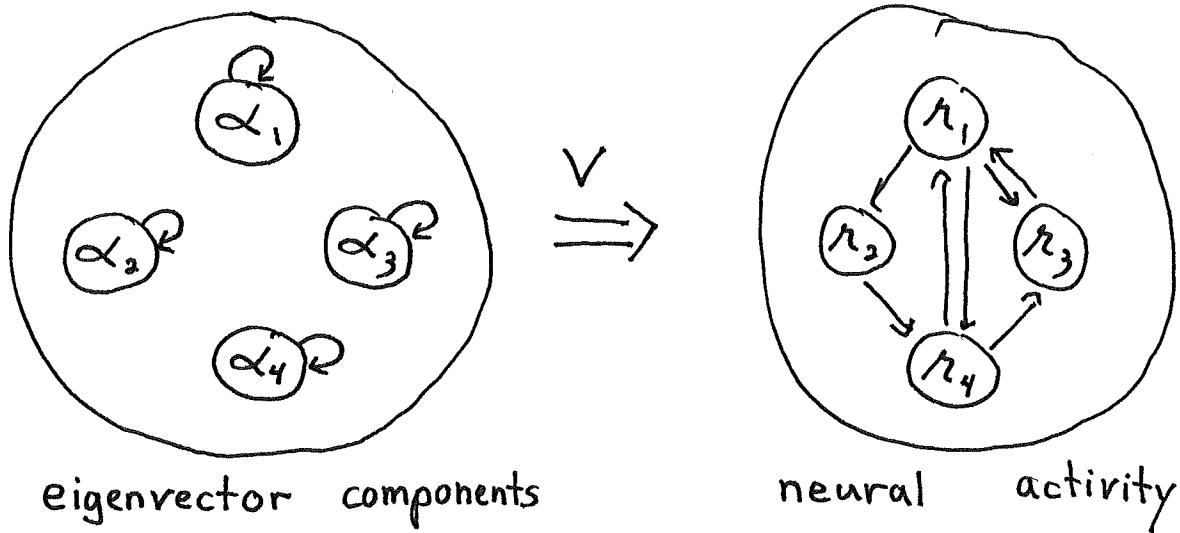
Targets



Similar PCA based analyses are very widespread in systems neuroscience.

PCA-like dimensionality reduction can become more motor cortex like by explicitly adding dynamics.

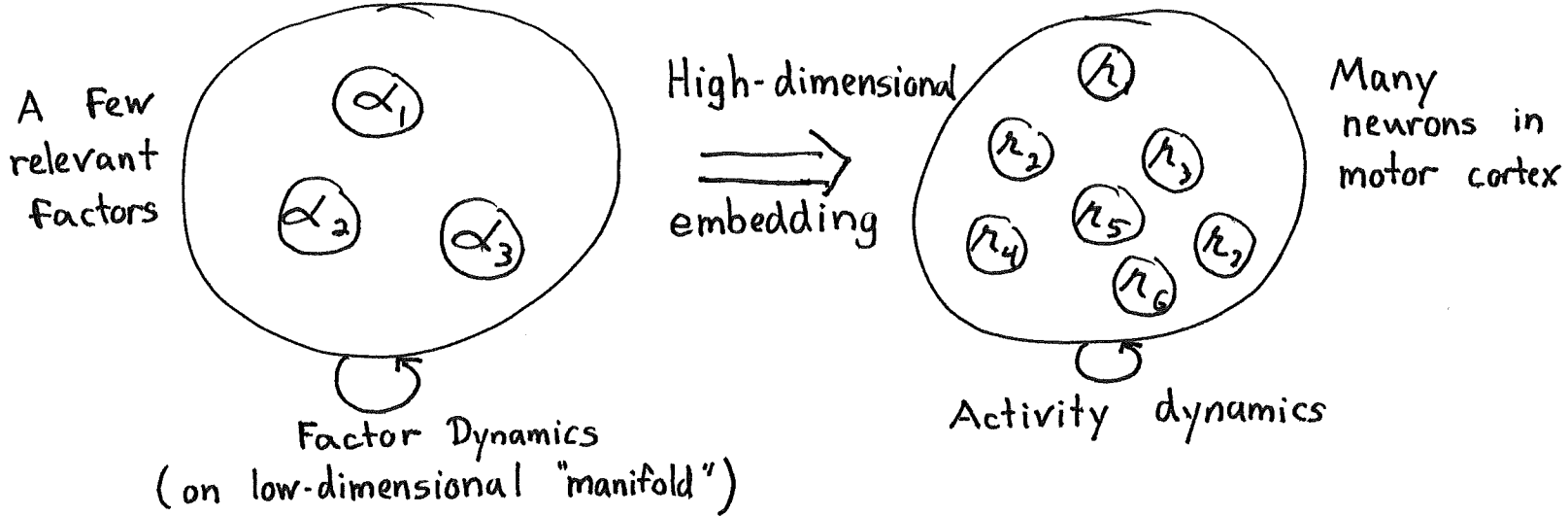
Linear neural network perspective from previous lectures



More general perspective:

Latent Factor Space

Neural Activity Space



Gaussian Process Factor Analysis (GPFA)

Probabilistic model linking latent space to activity

$$P(r_t | \alpha_t) = \mathcal{N} \left(\underbrace{C}_{\text{Coupling of neuron to mode}} \underbrace{\alpha_t}_{\text{Single timepoint, all neurons \& factors}} + \underbrace{d}_{\text{Offset of neuron}}, \underbrace{\Sigma}_{\text{Noise level of neuron}} \right)$$

Generative model for smooth neural trajectories:

$$P(\alpha_i) = \mathcal{N}(0, K_i)$$

Single factor, all time points Temporal covariance matrix.

$$K_{i,t_1,t_2} = \underbrace{\sigma_{n,i}^2}_{\text{White noise}} \delta_{t_1,t_2} + \underbrace{\sigma_{s,i}^2}_{\text{Signal strength}} e^{-\underbrace{(t_1-t_2)^2/(2\tau_i^2)}_{\text{controls correlation time}}}$$

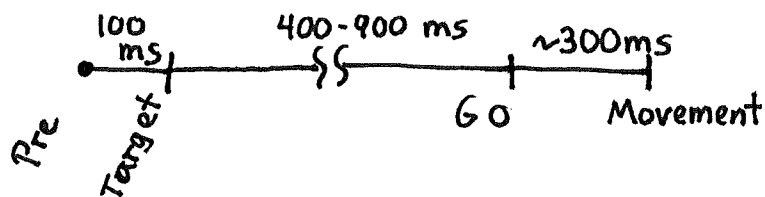
Overall scale of α is arbitrary due to C. Set,

$$\sigma_{n,i}^2 + \sigma_{s,i}^2 = 1$$

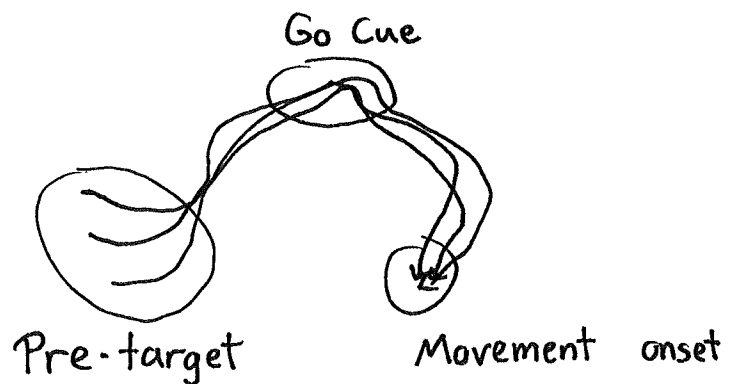
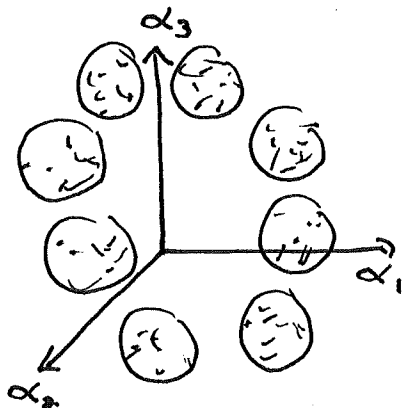
Fit model parameters with maximum likelihood

$$\ell(c, d, \Sigma, \sigma_n^2, \tau) = \log P(r | c, d, \Sigma, \sigma_n^2, \tau)$$

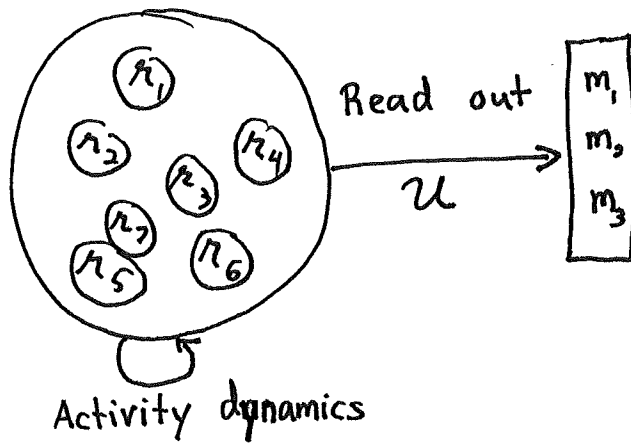
Applied to motor cortex during delayed reach task



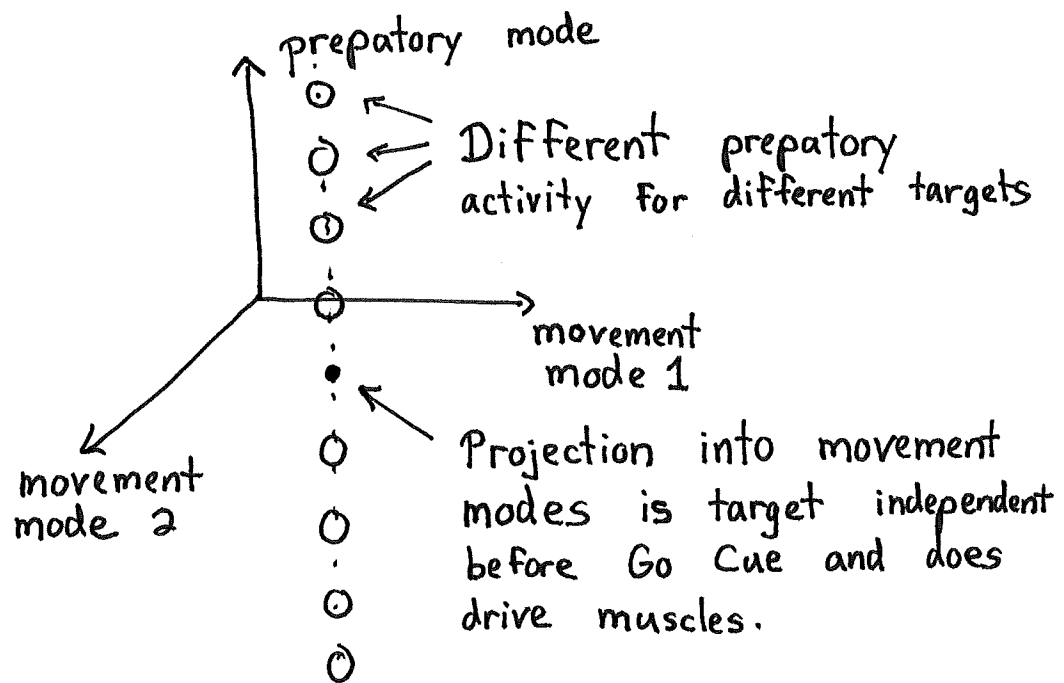
Factor analysis (i.e., GPFA with $\sigma_s^2 = 0$) reveals a low-dimensional space of preparatory activity, and GPFA constructs coherent trajectories throughout task.



Neural dynamics after target onset and before go cue lead to a tightening of neural trajectories and less trial-to-trial. Accordingly, movements are faster & more accurate. Preparatory activity can be conceptualized as "null space" activity that is not "read out" by muscles.



$$m = ur, \quad m(t) = 0 \Rightarrow (ur)(t) = 0 \not\Rightarrow r(t) = 0$$



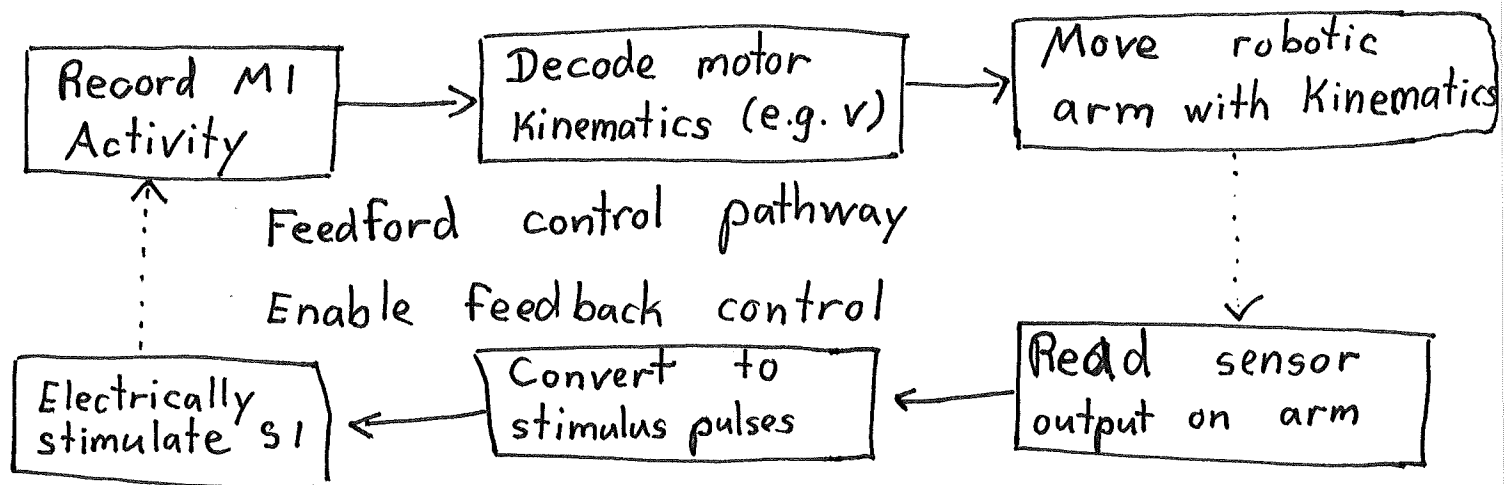
Simplest linear read out case.

$$[u] = N \times M, \quad M < N, \quad (N-M)\text{-dimensional null space with}$$

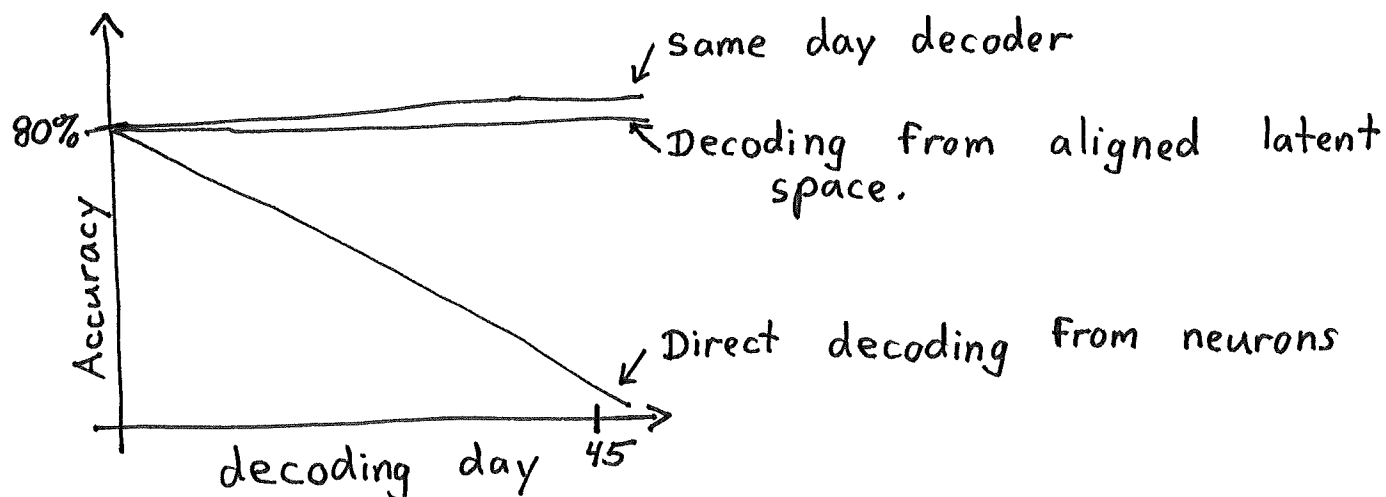
$$ur = 0$$

Neural decoding and Brain-Machine interfaces

The low-dimensional activity of neural activity has important implications. An important topic at the interface of computational neuroscience neuroscience and medicine is brain-machine interfaces, where one tries to control a prosthetic through brain activity. For example,



Such closed-loop control is being developed by Lee Miller at Northwestern and collaborators to restore sensorimotor function to paralyzed patients (Nature Reviews Neuroscience, 2014). Accurate decoding of motor kinematics is possible, despite unstable neural recordings, ~~impossible~~ for long time intervals because the latent factor dynamics are stable and discernable from relatively small neural populations (Gallego et al., Nature Neuroscience, 2020)



computational part due to Sara Solla (Northwestern).